

Line Tracking and Obstacle Avoidance Robot



Princess Sumaya جامعة
University الأميرة سميرة
for Technology للتكنولوجيا

Ismael. Al Hindi, Kamal. Alzaghah, Saleh. Al Saheb
Supervisor: Dr. Essam Al Qaralleh
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King Abdullah II School of Engineering
Princess Sumaya University for Technology

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Abstract

This project focuses on the design and implementation of an autonomous mobile robot capable of navigating a multi-zone track that includes line following, tunnel traversal, obstacle avoidance, bump handling, and precise parking. The robot uses three line-sensors, a light-dependent resistor (LDR), an ultrasonic distance sensor, two side infrared wall sensors, and a servo motor to operate independently without human input.

A three-loop control structure was developed to manage the robot's behavior across the different zones. This was achieved using timer-based interrupts, custom software-generated PWM, and direct register manipulation, in line with the course requirements. Testing showed that the robot was able to complete all sections of the track reliably, maintaining smooth motion, responsive control, and accurate parking at the final stage.

Introduction and Background

Autonomous mobile robots are widely used in applications such as intelligent transportation, inspection, and automation. The objective of this project is to design and implement an embedded autonomous robot capable of navigating a structured environment with minimal human intervention.

The challenge map consists of a sequence of zones, including start-up, line following, tunnel traversal, path navigation with obstacles, bump handling, and precision parking. Each zone presents distinct sensing and control challenges, requiring the robot to adapt its behavior based on environmental conditions. To meet these requirements, the system relies on discrete sensor interfaces, a standalone embedded controller, and low-level motor control implemented without the use of built-in libraries, in accordance with course constraints.

This project emphasizes robust decision-making under real-world limitations by focusing on:

- Reliable integration of multiple sensors for consistent navigation decisions
- Accurate timing control achieved without built-in delay functions
- A modular software structure separating sensing, control, and decision logic
- A simple and stable mechanical design focused on effective performance

Design

1. Mechanical Design

The robot uses a compact differential-drive chassis designed to meet size constraints and ensure stable operation.

The drive system consists of **two pairs of DC motors (four motors total)**, with two motors driving each side of the robot to increase torque and traction.

The four-motor configuration improves performance during obstacle avoidance, bump climbing, and sharp turns by reducing the risk of stalling.

Wheels are symmetrically mounted to maintain balance and enable smooth in-place rotation. Heavy components are positioned low on the chassis to keep the center of gravity low and improve stability during motion.

A servo-mounted flag is installed to visually indicate successful parking without interfering with movement.

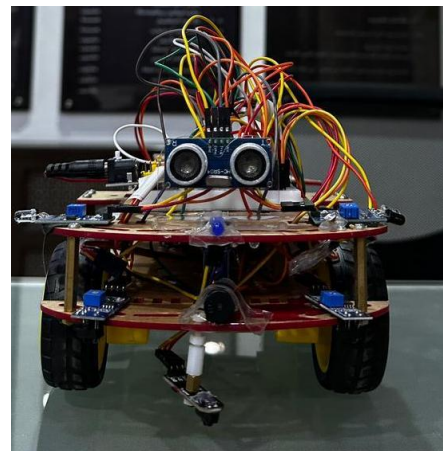
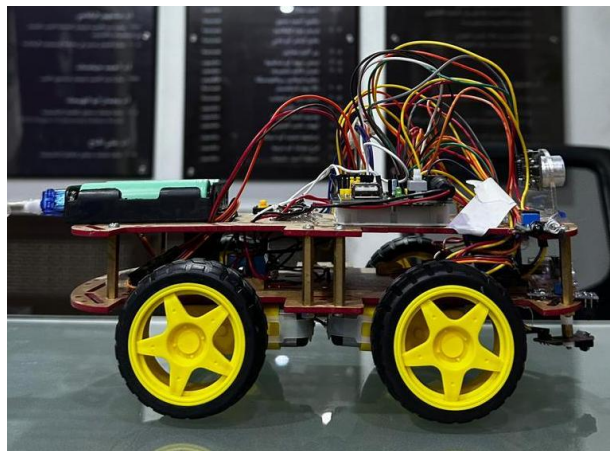
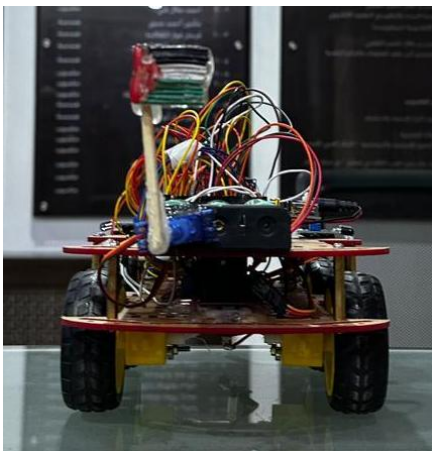


Figure 1: Photos of Mechanical Design

2. Electrical Design

The electrical system is centered around a PIC16F877A microcontroller operating at **8 MHz**, providing the timing base for interrupt scheduling, distance measurement, and software PWM. The robot integrates the following hardware modules:

- **Line Sensors (PORTB):** Three digital IR sensors are used for line following and parking-zone detection. The left and right sensors are connected to **RB6 and RB7**, while the middle sensor is connected to **RB5**.
- **LDR (AN0 / RA0):** A light-dependent resistor is interfaced through the ADC to detect low-light conditions in the tunnel. A calibrated threshold is used to reliably trigger tunnel entry/exit behavior.
- **Ultrasonic Sensor (PORTC + Timer1):** Distance measurement is performed using **RC4 (Trigger)** and **RC5 (Echo)**. The echo pulse width is measured using **Timer1**, then converted to distance for obstacle avoidance and parking.
- **Side IR Wall Sensors (RC6, RC7):** Two active-low wall sensors provide lateral feedback during path navigation to steer away from nearby walls.
- **Motor Driver and Outputs (PORTD):** Motor direction and enable signals are controlled using bit masks on PORTD, while **software-generated PWM** regulates speed. The **LED (RD4)** and **buzzer (RD5)** provide status indication, and a **servo on RC2** raises a flag after successful parking.
- **Power System:** A battery pack supplies the system through a regulator for stable logic voltage, with a shared ground between the microcontroller, sensors, and motor driver to ensure reliable operation.

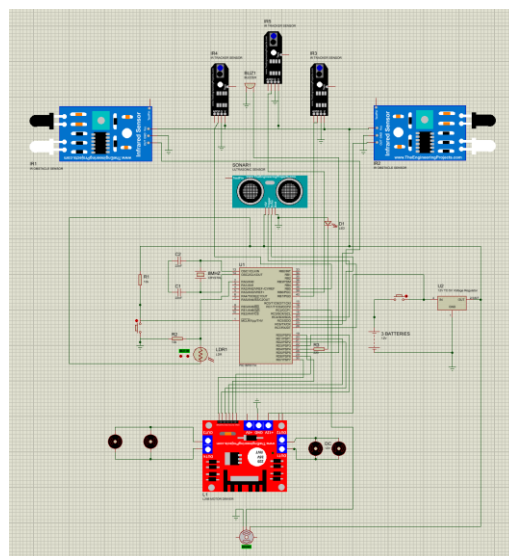


Figure 2: Hardware design

3. Software Design

The software is structured into three operational loops, reflecting the map stages:

Loop 1 – Line Following

- The robot uses three digital sensors to follow the black line.
- A Timer0 interrupt is used to achieve a startup delay of precisely three seconds.
- Light levels are continuously monitored by the LDR.
- The buzzer indicates tunnel entry when light intensity falls below a predetermined level.
- Until the robot leaves the tunnel, the buzzer stays on.

Loop 2 – Obstacle Avoidance & Path Navigation

- After leaving the tunnel, the line following is turned off.
- Frontal obstacles are detected by ultrasonic distance sensing.
- To prevent recurring collisions, a counter-based decision process alternates turning direction.
- For smoother navigation, side infrared sensors provide wall-following adjustments.
- A bump is correctly identified by continued forward motion rather than stopping.

Loop 3 – Parking

- The parking zone is detected when all line sensors read black.
- The robot slows down and approaches the wall using short PWM pulses for precision.
- A periodic buzzer sound is generated every second during parking.
- Once the target distance is reached, the robot performs a controlled 180-degree turn.
- Motors are stopped, the LED is turned off, and the servo raises the flag to indicate successful parking.

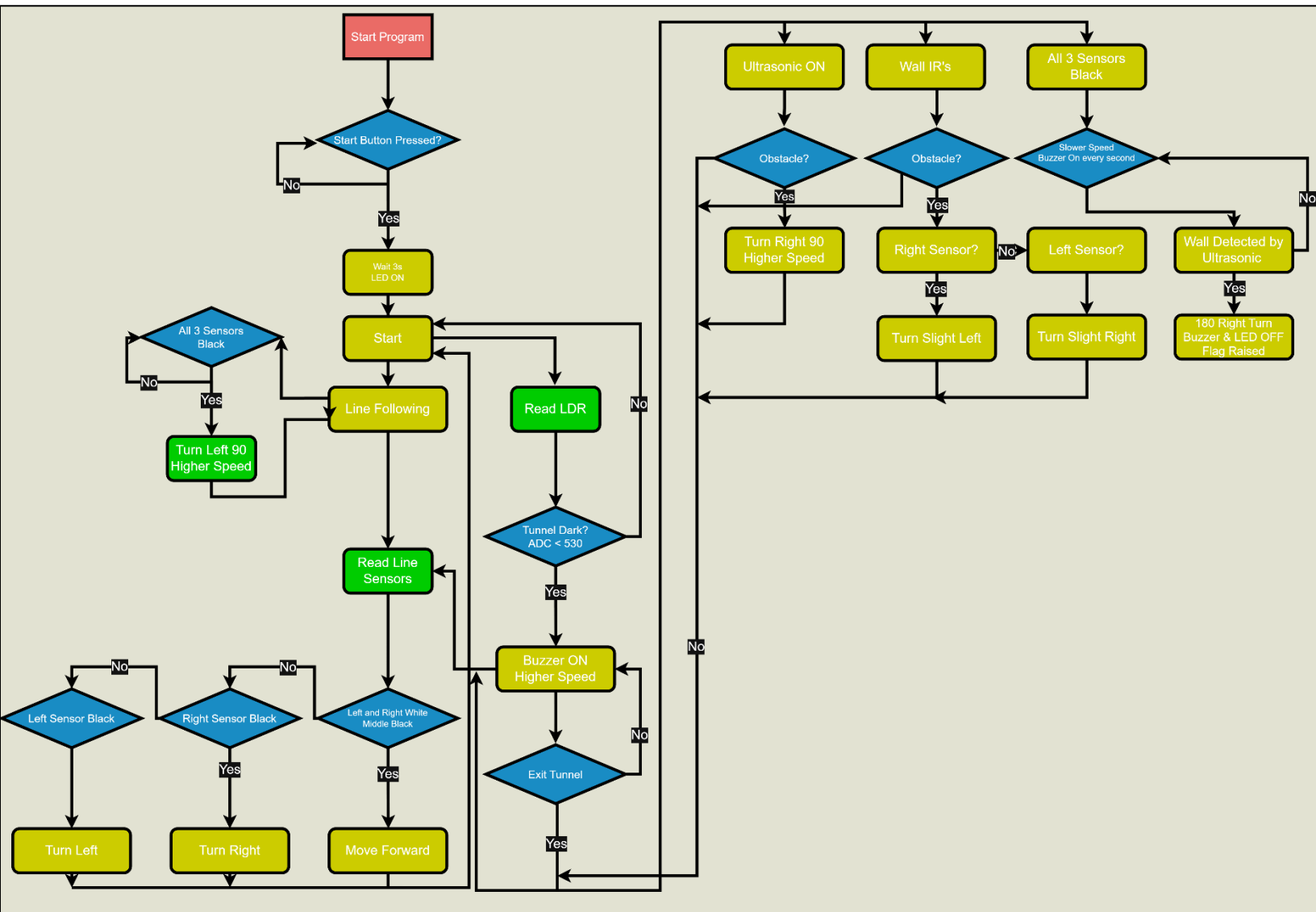


Figure 3: Software design

Problems

1. Line-following instability at higher speeds: Because of motor inertia and slow sensor reaction times, the robot periodically overshoot curves and junctions at higher PWM values.
2. Ultrasonic distance variations close to walls: The ultrasonic sensor occasionally gave erratic readings close to corners or angled walls during obstacle avoidance and parking, which affected stopping accuracy.
3. Turning accuracy variation: Due to the time-based nature of turning angles, minute changes in surface friction and battery voltage resulted in minor irregularities in 90° and 180° turns.

Recommendations

1. Reduce speed in crucial areas: Stability and control are enhanced by lowering PWM during curves, junctions, and parking.
2. Stabilize ultrasonic measurements: You can decrease false detections and increase parking accuracy by taking several distance readings before deciding.
3. Improve turn timing: Turning accuracy could be increased by adjusting turn delays under typical operating conditions or, if available, adding encoder feedback.

Conclusion

The design and execution of a completely autonomous mobile robot that can navigate a challenging multi-zone environment is successfully demonstrated by this project. Using a standalone embedded controller, the system combines wall following, obstacle avoidance, low-light recognition, line-following, and precision parking. Without using built-in libraries, the robot accomplishes accurate sensor-driven decision making and dependable motion control through direct manipulation of microcontroller registers, timer-based interrupts, and manually constructed PWM control.

Experimental testing verifies that the robot responds appropriately to environmental changes, parks neatly within the allocated area, and seamlessly moves between all phases of the track while offering visual and aural feedback. In addition to achieving the project's goals, this work upholds fundamental embedded systems concepts like low-level timing management, hardware–software integration, and real-time control. The system has a solid basis for more sophisticated robotic applications because of its modular design, which permits future development, such as autonomous mapping or closed-loop motor control.